Math for CS & AI: Homework 9

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Problem 4.18

Let $X_1, \ldots, X_n$ be i.i.d. random variables drawn from a distribution with mean μ and variance σ^2. Consider the following limit for every $\alpha \in \mathbb{R}$:

$$\lim_{n\to\infty} K_{Z_\alpha^{(n)}}(t), \quad \text{where } Z_\alpha^{(n)} := \frac{1}{n^\alpha} \sum_{i=1}^{n} (X_i - \mu).$$

Determine the range of α for which the limit exists and is finite.

Answer:

Let $K(t)$ be the cumulant generating function of X_i. By the properties of CGFs, we have:

$$K_{Z_\alpha^{(n)}}(t) = nK(tn^{-\alpha}) - n^{1-\alpha}\mu t.$$

We expand $K(s)$ around 0 and obtain

$$K(s) = \mu s + \frac{\sigma^2}{2}s^2 + \mathcal{O}(s^3).$$

Substituting $s = tn^{-\alpha}$, for every fixed $t \in \mathbb{R}$, as $n \to \infty$, we have

$$\begin{aligned} K_{Z_\alpha^{(n)}}(t) &= n\left(\mu t n^{-\alpha} + \frac{\sigma^2}{2} \cdot t^2 n^{-2\alpha} + \mathcal{O}(n^{-3\alpha})\right) - n^{1-\alpha}\mu t \\ &= \frac{\sigma^2}{2} \cdot t^2 n^{1-2\alpha} + \mathcal{O}(n^{1-3\alpha}). \end{aligned}$$

Therefore, as $n \to \infty$, this expression converges to the following limit:

$$K_{Z_\alpha^{(n)}}(t) \to \begin{cases} \frac{1}{2}\sigma^2 t^2, & \alpha = 1/2, \\ 0, & \alpha > 1/2. \end{cases}$$

For $\alpha < 1/2$, the term $n^{1-2\alpha}$ diverges to infinity as $n \to \infty$, so the limit does not exist. Thus, the limit exists and is finite if and only if $\alpha \geq 1/2$.

Problem 4.19

Let $X^{(n)} \sim \text{Binomial}(n, p_n)$, where $p_n = \lambda/n$ for some fixed $\lambda > 0$. Directly compute the following two limits and show that they are the same:

(a) $\lim_{n\to\infty} K_{X^{(n)}}(t)$;

(b) $\log\left(\sum_{k\geq 0} e^{kt} \lim_{n\to\infty} \Pr\left[X^{(n)} = k\right]\right)$.

Note: The limiting distribution is known as the Poisson distribution, which describes the number of rare events occurring in a large number of independent trials. See also Exercise 5.3.

Answer:

For part (a), the CGF of $X^{(n)} \sim \text{Binomial}(n, p_n)$ is $K_{X^{(n)}}(t) = n\log(1 - p_n + p_n e^t)$. Substituting $p_n = \lambda/n$, for any fixed $t \in \mathbb{R}$, as $n \to \infty$, we have

$$\begin{aligned}
\lim_{n\to\infty} K_{X^{(n)}}(t) &= \lim_{n\to\infty} n\log\left(1 + \frac{\lambda}{n}(e^t - 1)\right) \\
&= \lim_{n\to\infty} n\left(\frac{\lambda}{n}(e^t - 1) + \mathcal{O}(n^{-2})\right) \\
&= \lambda(e^t - 1).
\end{aligned}$$

For part (b), we have

$$\begin{aligned}
\log\left(\sum_{k\geq 0} e^{kt} \lim_{n\to\infty} \Pr\left[X^{(n)} = k\right]\right) &= \log\left(\sum_{k\geq 0} e^{kt} \lim_{n\to\infty} \binom{n}{k}\left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}\right) \\
&= \log\left(\sum_{k\geq 0} e^{kt} \lim_{n\to\infty} \frac{n^k}{k!} \cdot \frac{\lambda^k}{n^k} \cdot e^{-\lambda}\right) \\
&= \log\left(e^{-\lambda} \sum_{k\geq 0} \frac{(\lambda e^t)^k}{k!}\right) \\
&= \lambda(e^t - 1).
\end{aligned}$$

Therefore, the limits obtained in parts (a) and (b) are identical, as required.

Problem 5.1

Let A be a set that we do not know whether it is countable or uncountable. Prove the following statements:

(a) If there exists an injection $f : A \to B$ from A to a countable set B, then A is countable.

(b) If there exists a surjection $f : A \to B$ from A to an uncountable set B, then A is uncountable.

Here, we say that a function $f : A \to B$ is an injection if for every two distinct elements $a_1, a_2 \in A$, we have $f(a_1) \neq f(a_2)$. Similarly, we say that a function $f : A \to B$ is a surjection if for every element $b \in B$, there exists an element $a \in A$ such that $f(a) = b$.

Answer:

For part (a), since B is a countable set, by definition there exists an injection $g : B \to \mathbb{N}$. Define the function $h : A \to \mathbb{N}$ by $h(x) = g(f(x))$. Since the composition of two injective functions is injective, h is an injection from A to $\mathbb{N}$. Therefore, A is countable.

For part (b), we proceed by contradiction. Assume that A is countable. Since $f : A \to B$ is a surjection, for every $b \in B$, the preimage $f^{-1}(\{b\}) \subseteq A$ is non-empty. We can therefore define a function $g : B \to A$ by choosing, for each $b \in B$, an arbitrary element $g(b) \in f^{-1}(\{b\})$. We now show that g is injective: if $g(b_1) = g(b_2) = a$ for some $a \in A$, then $f(a) = b_1$ and $f(a) = b_2$, so $b_1 = b_2$. Thus g is an injection from B to A. By part (a), this implies that B is countable, which contradicts the assumption that B is uncountable. Therefore, A must be uncountable.

Problem 5.2

For each of the following sets, determine whether it is countable or uncountable:

(a) The set of all finite tuples of integers: $\bigcup_{k \geq 1} \mathbb{Z}^k$.

(b) The set of all Python code that can be executed. You can view a Python code as a finite sequence of symbols from a finite alphabet. Some sequences are executable, some are not.

(c) The set of all functions $f : \mathbb{N} \to \{0, 1\}$.

(d) The set of functions $f : \mathbb{N} \to \{0, 1\}$ that can be implemented by a Python program. Here a program implements a function f if for every input $n \in \mathbb{N}$, the program outputs the value of $f(n)$.

Note: The functions in the last set are called computable functions. For anyone interested in learning more about them, take a course on the theory of computation!

Answer:

For part (a), the set is countable. We can enumerate the elements of this set by first ordering them by the sum of the absolute values of their elements, and then by lexicographical order. This enumeration is valid because there are only finitely many tuples with a given sum of absolute values, so every tuple has only finitely many predecessors.

For part (b), the set is countable. We can encode each symbol from the finite alphabet into a unique integer, so every finite sequence of symbols corresponds to a finite tuple of integers. Thus the set of all executable Python code is a subset of the set considered in part (a), which is countable by the result from Problem 5.1(a).

For part (c), the set is uncountable. Assume for contradiction that we can enumerate all such functions as $f_1, f_2, f_3, \ldots$, where each $f_i : \mathbb{N} \to \{0, 1\}$. Then define a new function $g : \mathbb{N} \to \{0, 1\}$ by $g(n) = 1 - f_n(n)$ for all $n \in \mathbb{N}$. By construction, g differs from every f_i at position i, so g is not in the enumeration. This leads to a contradiction. Therefore, this set is uncountable.

For part (d), the set is countable. Every computable function is implemented by at least one Python program, and the set of all Python programs is a subset of the set of all finite sequences of symbols from a finite alphabet. By part (b), this set is countable. Therefore, the set of computable functions, being the image of a countable set under the "implements relation", is also countable.

Problem 5.3

The Poisson distribution with parameter $\lambda > 0$, denoted by Poisson(λ), is a discrete probability distribution with probability mass function

$$P(k) := \frac{\lambda^k}{k!} e^{-\lambda}, \quad \text{for } k \in \mathbb{N}.$$

(a) Prove that P is a valid probability mass function. That is, prove that $\sum_{k \geq 0} P(k) = 1$.

(b) Let $X \sim \text{Poisson}(\lambda)$. Determine $\mathbb{E}[X]$.

(c) Let $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ be independent Poisson random variables. Prove that $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Answer:

For part (a), we have $e^\lambda = \sum_{k \geq 0} \frac{\lambda^k}{k!}$ for all $\lambda \in \mathbb{R}$. This implies that

$$\sum_{k \geq 0} P(k) = e^{-\lambda} \sum_{k \geq 0} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1.$$

For part (b), we have

$$\mathbb{E}[X] = e^{-\lambda} \sum_{k \geq 0} \frac{\lambda^k}{k!} \cdot k = \lambda e^{-\lambda} \sum_{k \geq 1} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda.$$

For part (c), for any non-negative integer n, by the independence of X_1 and X_2, we have

$$\begin{aligned}
\Pr[X_1 + X_2 = n] &= \sum_{i=0}^{n} \Pr[X_1 = i] \Pr[X_2 = n - i] \\
&= \sum_{i=0}^{n} \frac{\lambda_1^i}{i!} e^{-\lambda_1} \cdot \frac{\lambda_2^{n-i}}{(n-i)!} e^{-\lambda_2} \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{n!} \sum_{i=0}^{n} \binom{n}{i} \lambda_1^i \lambda_2^{n-i} \\
&= e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}.
\end{aligned}$$

This implies $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

Problem 5.4

The exponential distribution with parameter $\lambda > 0$, denoted by $\text{Exp}(\lambda)$, is a continuous probability distribution with probability density function

$$f(x) := \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

(a) Prove that f is a valid probability density function. That is, prove that $\int_{-\infty}^{\infty} f(x)dx = 1$.

(b) Let $X \sim \text{Exp}(\lambda)$. Determine $\mathbb{E}[X]$.

(c) Determine the MGF and CGF of $\text{Exp}(\lambda)$.

(d) Let $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ be independent exponential random variables. Prove that $X_1 + X_2$ does not follow any exponential distribution.

Answer:

For part (a), we have

$$\int_0^{+\infty} \lambda e^{-\lambda x} \mathrm{d}x = -\int_0^{+\infty} e^{-\lambda x} \mathrm{d}(-\lambda x) = -e^{-\lambda x}\Big|_0^{+\infty} = 1.$$

For part (b), we have

$$\mathbb{E}[X] = \int_0^{+\infty} \lambda x e^{-\lambda x} \mathrm{d}x = -\left(x e^{-\lambda x}\Big|_0^{+\infty} - \int_0^{+\infty} e^{-\lambda x} \mathrm{d}x \right) = \frac{1}{\lambda}.$$

For part (c), let $M(t)$ and $K(t)$ denote the MGF and CGF of $X \sim \text{Exp}(\lambda)$, respectively. We have:

$$M(t) = \mathbb{E}[e^{tX}] = \int_0^{+\infty} \lambda e^{-\lambda x} \cdot e^{tx} \mathrm{d}x = \frac{\lambda}{\lambda - t}, \quad \text{for } t < \lambda,$$
$$K(t) = \log M(t) = \log\left(\frac{\lambda}{\lambda - t}\right), \quad \text{for } t < \lambda.$$

For part (d), for any $x \geq 0$, by the independence of X_1 and X_2, the probability density function of $X_1 + X_2$ is given by the convolution:

$$\begin{aligned} f_{X_1+X_2}(x) &= \int_0^x f_{X_1}(t) f_{X_2}(x-t) \mathrm{d}t \\ &= \int_0^x \lambda_1 e^{-\lambda_1 t} \cdot \lambda_2 e^{-\lambda_2 (x-t)} \mathrm{d}t \\ &= \lambda_1 \lambda_2 e^{-\lambda_2 x} \int_0^x e^{(\lambda_2 - \lambda_1)t} \mathrm{d}t. \end{aligned}$$

We consider two cases:

- If $\lambda_1 \neq \lambda_2$, then

$$f_{X_1+X_2}(x) = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} \left(e^{-\lambda_1 x} - e^{-\lambda_2 x} \right).$$

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- If $\lambda_1 = \lambda_2 = \lambda$, then
$$f_{X_1+X_2}(x) = \lambda^2 e^{-\lambda x} \int_0^x \mathrm{d}t = \lambda^2 x e^{-\lambda x}.$$

In both cases, the probability density function cannot be written in the form of the PDF of an exponential distribution. Therefore, $X_1 + X_2$ does not follow any exponential distribution.

Problem 5.5

For any probability distribution $\mathcal{D}$ on $\mathbb{R}$, define the cumulative distribution function (CDF) of $\mathcal{D}$ as

$$F_{\mathcal{D}}(x) := \mathcal{D}((-\infty, x]), \quad \text{for } x \in \mathbb{R}.$$

If $X \sim \mathcal{D}$, then $F_{\mathcal{D}}(x) = \Pr[X \leq x]$, which is the probability that X is less than or equal to x. It can be shown that the CDF is a non-decreasing function: $F_{\mathcal{D}}(x) \leq F_{\mathcal{D}}(y)$ for all $x \leq y$. Another good property is that the CDF is always right-continuous: $\lim_{x \to x_0^+} F_{\mathcal{D}}(x) = F_{\mathcal{D}}(x_0)$ for all $x_0 \in \mathbb{R}$.

Define the quantile function of $\mathcal{D}$ as

$$F_{\mathcal{D}}^{-1}(p) := \inf \{x \in \mathbb{R} : F_{\mathcal{D}}(x) \geq p\}, \quad \text{for } p \in [0, 1].$$

In other words, the quantile function gives a threshold value x that tries its best to separate the probability mass of X into a lower part of mass p and an upper part of mass $1 - p$.

(a) Determine the CDF and quantile function of the exponential distribution $\mathrm{Exp}(\lambda)$.

(b) Determine the CDF and quantile function of the geometric distribution $\mathrm{Geometric}(p)$ (Example 3.26).

Answer:

For part (a), the CDF and quantile function of $\mathrm{Exp}(\lambda)$ are

$$F_{\mathrm{Exp}(\lambda)}(x) = \begin{cases} 0, & x < 0, \\ \int_0^x \lambda e^{-\lambda t} \mathrm{d}t = 1 - e^{-\lambda x}, & x \geq 0, \end{cases}$$
$$F_{\mathrm{Exp}(\lambda)}^{-1}(p) = \inf \left\{x \in \mathbb{R} : 1 - e^{-\lambda x} \geq p\right\} = -\frac{\ln(1-p)}{\lambda}.$$

For part (b), the CDF and quantile function of $\mathrm{Geometric}(p)$ are

$$F_{\mathrm{Geometric}(p)}(x) = \begin{cases} 0, & x < 1, \\ \sum_{k=1}^{\lfloor x \rfloor} (1-p)^{k-1} p = 1 - (1-p)^{\lfloor x \rfloor}, & x \geq 1, \end{cases}$$
$$F_{\mathrm{Geometric}(p)}^{-1}(p_0) = \inf \left\{x \in \mathbb{R} : 1 - (1-p)^{\lfloor x \rfloor} \geq p_0\right\} = \left\lceil \frac{\ln(1-p_0)}{\ln(1-p)} \right\rceil.$$

Problem 5.6

Let $\mathcal{D}$ be a probability distribution on $\mathbb{R}$. If it is not possible to generate samples from $\mathcal{D}$ in a computer program, then $\mathcal{D}$ would be meaningless for computer scientists. It is easy to generate samples from the uniform distribution on $[0, 1)$, which we denote by U for short. For example, in Python, we can write

```
import random
print(random.random())
```

To generate samples from an arbitrary distribution $\mathcal{D}$, we have to apply a deterministic transformation to the uniform samples.

(a) Let $Z \sim U$. Determine the probability density function of $\alpha Z + \beta$ for any $\alpha \in \mathbb{R} \setminus \{0\}$ and $\beta \in \mathbb{R}$.

(b) Let $Z \sim U$ and $F_{\mathcal{D}}^{-1}$ be the quantile function of $\mathcal{D}$. Show that

$$F_{\mathcal{D}}^{-1}(Z) \sim \mathcal{D}.$$

This means that we can generate samples from $\mathcal{D}$ by generating samples from U and then applying the quantile function of $\mathcal{D}$ to them.

Answer:

For part (a), the PDF of $Z \sim U$ is $f_Z(x) = \mathbb{1}_{[0 \le x < 1]}$. By the change of variables formula for probability density functions, we obtain

$$f_{\alpha Z+\beta}(x) = \frac{1}{|\alpha|} f_Z\left(\frac{x-\beta}{\alpha}\right) = \frac{1}{|\alpha|}\mathbb{1}_{[0 \le \frac{x-\beta}{\alpha} < 1]}.$$

For part (b), let $F = F_{\mathcal{D}}$ be the CDF of $\mathcal{D}$. By definition, the quantile function is $F^{-1}(p) = \inf\{x \in \mathbb{R} : F(x) \ge p\}$.

Now consider the CDF of the random variable $F^{-1}(Z)$. For any $x_0 \in \mathbb{R}$, we have

$$\begin{aligned} F_{F^{-1}(Z)}(x_0) &= \Pr\left[F^{-1}(Z) \le x_0\right] \\ &= \Pr\left[\inf\{x \in \mathbb{R} : F(x) \ge Z\} \le x_0\right]. \end{aligned}$$

Since F is non-decreasing and right-continuous, the event $\inf\{x : F(x) \ge Z\} \le x_0$ is equivalent to $F(x_0) \ge Z$. Therefore,

$$\begin{aligned} F_{F^{-1}(Z)}(x_0) &= \Pr\left[Z \le F(x_0)\right] \\ &= F(x_0), \end{aligned}$$

where the last equality follows from the fact that $Z \sim U[0, 1)$.

Thus, the CDFs of $\mathcal{D}$ and $F^{-1}(Z)$ are identical, which implies $F_{\mathcal{D}}^{-1}(Z) \sim \mathcal{D}$.

Problem 5.7

Let $X \sim \mathcal{N}(0, \sigma^2)$. Compute the moments $\mathbb{E}[X^k]$ for all integers $k \geq 1$.

Answer:

For a Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$, the MGF of X is

$$M_X(t) = e^{\sigma^2 t^2/2} = \sum_{k \geq 0} \frac{\sigma^{2k} t^{2k}}{2^k k!}.$$

Recall that the MGF has the general expansion $M_X(t) = \sum_{k \geq 0} \frac{\mathbb{E}[X^k]}{k!} t^k$. Comparing the coefficients of t^k in the two expansions, we obtain

$$\mathbb{E}[X^k] = \mathbb{1}_{[2|k]} \cdot \frac{\sigma^k k!}{2^{k/2}(k/2)!}.$$

Problem 5.8

Define $\bar{\Phi}(x) := \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \mathrm{d}u$ as the tail probability of the standard Gaussian distribution. Prove that

$$\bar{\Phi}(x) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-x^2/2} - \frac{1}{x^3} e^{-x^2/2} + \int_x^{+\infty} \frac{3}{u^4} e^{-u^2/2} \mathrm{d}u \right).$$

Note: This explains why $\bar{\Phi}(x) \sim \frac{1}{\sqrt{2\pi}} \frac{e^{-x^2/2}}{x}$ in Theorem 5.28.

Answer:

We use integration by parts repeatedly to obtain the following identity:

$$\begin{aligned}
\bar{\Phi}(x) &= \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \mathrm{d}u \\
&= \frac{1}{\sqrt{2\pi}} \left(-\frac{1}{u} e^{-u^2/2} \Big|_x^{+\infty} - \int_x^{+\infty} \frac{1}{u^2} e^{-u^2/2} \mathrm{d}u \right) \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-x^2/2} - \int_x^{+\infty} \frac{1}{u^2} e^{-u^2/2} \mathrm{d}u \right) \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-x^2/2} - \left(-\frac{1}{u^3} e^{-u^2/2} \Big|_x^{+\infty} - \int_x^{+\infty} \frac{3}{u^4} e^{-u^2/2} \mathrm{d}u \right) \right) \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{x} e^{-x^2/2} - \frac{1}{x^3} e^{-x^2/2} + \int_x^{+\infty} \frac{3}{u^4} e^{-u^2/2} \mathrm{d}u \right).
\end{aligned}$$